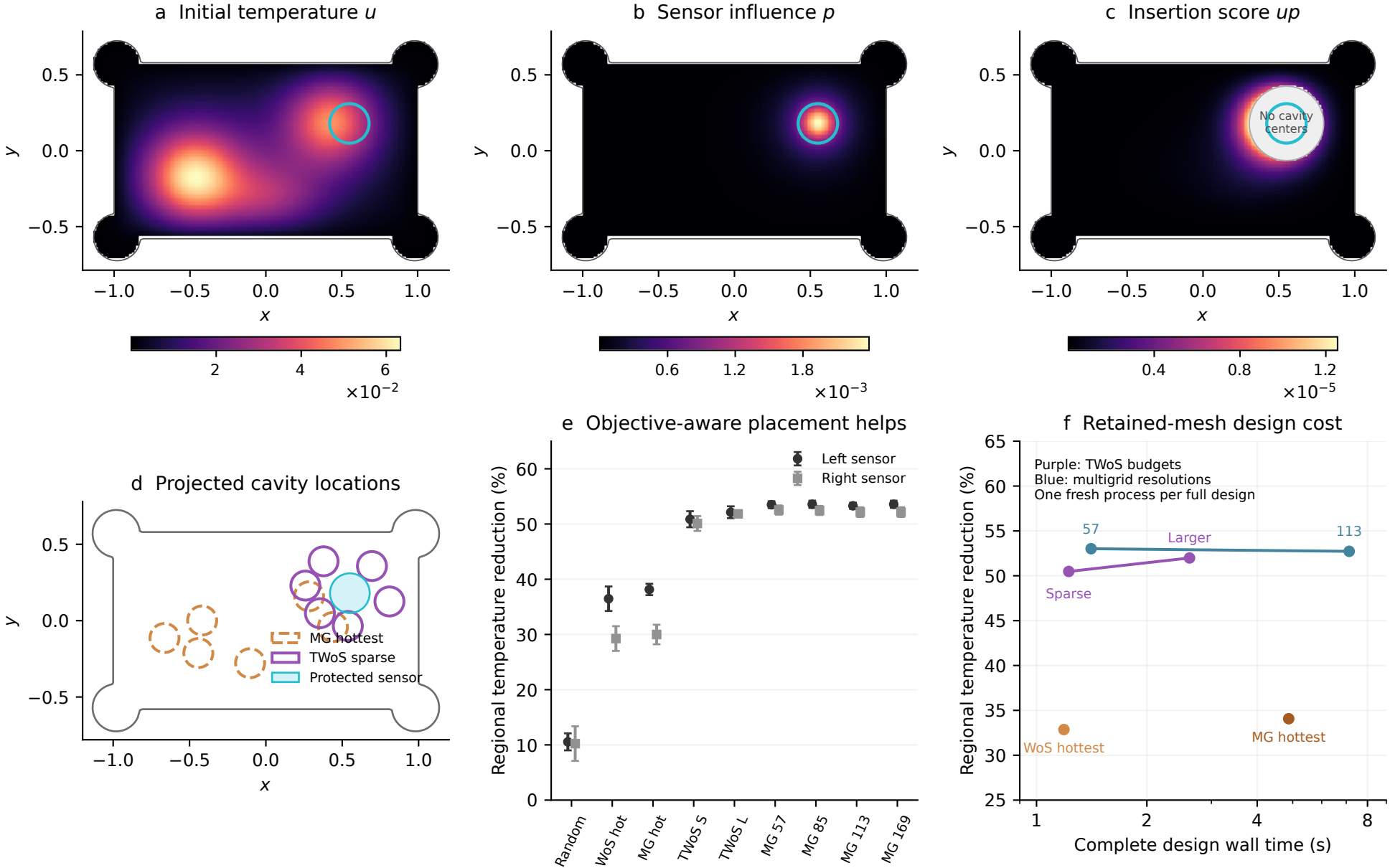

a Initial temperature u
b Sensor influence p
c Insertion score up
No cavity centers
d Projected cavity locations
MG hottest
TWoS sparse
Protected sensor
e Objective-aware placement helps
Left sensor
Right sensor
Regional temperature reduction (%)
Random
WoS hot
MG hot
TWoS S
TWoS L
MG 57
MG 85
MG 113
MG 169
f Retained-mesh design cost
Purple: TWoS budgets
Blue: multigrid resolutions
One fresh process per full design
57
Larger
113
Sparse
WoS hottest
MG hottest
Complete design wall time (s)
x
y